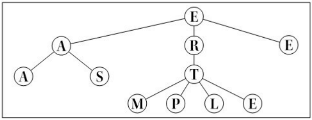

Figure 2: A SAMPLE TREE

```
#include <stdlib.h>
#include <stdio.h>
#include <math.h>
#include <string.h>
#define STD 23742246
main(int argc, char* argv)
{
  int no=0;
  double x,y;
  int i,count=0;
  double z;
  double pi;
  printf("Enter the number of iterations to be done to estimate the value of
pi: ");
  scanf("%d",&no);
  srand(STD);
  count=0;
  for ( i=0; i<no; i++) {
    x = (double)rand()/RAND_MAX;
    y = (double)rand()/RAND_MAX;
    z = x*x+y*y;
    if (z<=1) count++;
    }
  pi=(double)count/no*4;
  printf("No of trials= %d , The Estimated value of pi is %g \n",no,pi);
}
```

You can see that the estimated value of Pi approaches the correct value when the number of steps in the procedure increases (dynamically). If you look at the code carefully, you will see that the code terminates after a finite period (provided the input is finite!):

```
aasisvinayak@GNU-BOX:~/Documents/Desktop$ ./pi
Enter the number of iterations to be done to estimate the value of pi: 44
No of trials= 44 , The Estimated value of pi is 2.81818
aasisvinayak@GNU-BOX:~/Documents/Desktop$ ./pi
Enter the number of iterations to be done to estimate the value of pi: 1234567
No of trials= 1234567 , The Estimated value of pi is 3.14508
```

We have seen the implemented codes. Now, try writing the algorithms behind the codes. As you have the code, it is quite easy!

Another way of visualising our algorithms is the Tree Model. Beginners can opt for this since this will never cause any bewilderment. Figure 2 shows a model of A SAMPLE TREE. This will lend a hand to such users to comprehend the idea of branching.

Today's problem

We have a theorem: If Ω be the set of all exact execution paths for a defined 'A' on input x, then...

$$\sum_{\lambda \in \Omega} 2^{-|\lambda|} \leq 1$$

Can you prove this? (Note that all symbols have the usual meaning that we used for other algorithms/ problems.)

Clues:
Start with...

$$\alpha_l := \sum_{\lambda \in \Omega} 2^{-|\lambda|} \leq 1$$

...and deduce

$$\alpha_l \leq 1$$

Then, try to get...

$$\sum_{\lambda \in \Omega} 2^{-|\lambda|} = \lim_{l \to \infty} \alpha_l \leq 1$$

...by considering ...

$$\alpha_l = 2^{-l} \left| S_k \right|$$

If you can reach this juncture, the rest will follow. For all these, you can assume that A halts with probability α on input x. If $\alpha = 1$, we define the distribution $P: \Omega \to [0, 1]$.

Beginners need not worry much. We will be dealing with complex problems much later in our Voyage!

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